

When will Auracast be available?

If you are rocking Bluetooth devices with versions 5.2 or 5.3 and LE Audio, you are in luck. As part of the Bluetooth family, Auracast doesn't require extra tech wizardry to join the party—it just needs your devices to be up to date. The ball is then in the court of individual companies, and the buzz at CES and throughout 2023 suggests that many audio brands are gearing up to roll out support for Auracast via over-the-air updates in 2024.

places into communal hubs, making the audio experiences more engaging and shared. No more isolation from the world around us; instead, Auracast utilises Bluetooth Low Energy (LE) to create environments where audio sharing is seamless.

In crowded places with silent TVs, like sports bars or exhibitions, how do you tune into your preferred audio without the hassle? Auracast allows a single audio source to broadcast to an unlimited number of listening devices without the need for individual connections. It is a breakthrough that opens a world of possibilities—private listening, staying informed with announcements, or simply sharing music seamlessly with friends. Auracast, supported by Bluetooth LE audio, is shaping the future of how we interact with sound.

As a user, you might wonder, “How can I join Auracast?” Well, it is a breeze! First, you kick off with a familiar searching process, much like the routine of connecting to Wi-Fi—it's intuitive and comfortable. A simple QR code scan is all it takes to start the Auracast broadcasts, levelling up convenience and connectivity. Next, you dive into Tapping, where a singular tap serves as your golden ticket to quick and hassle-free access, reflecting the streamlined efficiency we have come to love in modern tap-to-pay systems. With these three steps—searching, scanning, and tapping—Auracast unfolds as an exciting and user-friendly journey, promising a seamless entry into a world of groundbreaking audio experiences that await at your fingertips.

In the Auracast ecosystem, we simplify roles using generic names. When we talk about the role of receiving audio broadcasts, it goes by the name ‘Auracast receiver,’ encompassing titles like broadcast sink, public broadcast sink, acceptor, hearing aid, and broadcast media receiver. On the flip side, the term “Auracast transmitter” takes the lead when it comes to sending out audio broadcasts, embracing roles such as broadcast source, public broadcast source, initiator, and broadcast media sender.

Simple Auracast transmitters

The physical form of an Auracast transmitter does not dictate its function. It could be a plug-in dongle for devices like phones, tablets, or PCs, a small box connected to sound systems or TVs, or even seamlessly integrated as a system-on-chip (SoC) or embedded module within consumer electronics. The versatility in form ensures that Auracast functionality can be effortlessly incorporated into various devices, promising a standardised audio experience for the users, regardless of the technical complexities behind the scenes.

Ensuring compliance with Auracast trademarks, all transmitters must transmit broadcast audio at 16kHz or 24kHz using the LC3 codec. This ensures compatibility with a wide range of devices, including resource-constrained ones like hearing aids, which may only support a standard quality public broadcast audio stream.

Auracast transmitters designed for public spaces are required to default to broadcasting a stand-

ard quality public broadcast audio stream. This ensures compatibility with various receiver devices, especially those with limited resources like hearing aids, which may only support standard quality streams. By prioritising standard quality in public locations, the smaller audio stream packets offer improved range and robustness against interference compared to high quality packets.

On the other hand, personal Auracast transmitters may default to high quality audio streams but must allow users to choose standard quality transmission. The manufacturer, and potentially the user, decide whether the standard quality stream is transmitted alongside or instead of the high quality stream, providing flexibility in audio preferences.

Bluetooth LE Audio broadcast-ers utilise extended and periodic advertisements to convey information about broadcast streams. Auracast transmitters, following the PBP specification, include the public broadcast announcement within extended advertisements. This announcement informs devices about the audio broadcast, helping them determine if it contains a standard quality or high-quality audio stream. This assists devices in selectively scanning for relevant streams, enhancing efficiency.

To ensure a user-friendly experience, Auracast transmitters include human-readable text strings in extended and periodic advertising data. Metadata relevant to all broadcast audio streams is included in extended advertising data, while specific or regularly changing metadata is placed in periodic advertising data. These configurations aim to identify streams, providing users with clear information unambiguously. The recommended settings for a public Auracast transmitter broadcasting a single broadcast is ochronous.